Practice Quiz: Cognition — Mental Models of the Finished Piece

Part A: Multiple Choice

1. Visualizing the finished piece from written instructions is best described as: A) Guessing what it will look like B) Running a generative model simulation forward C) Copying from a photo D) Unnecessary for crochet

2. When you instantly decode "dc" into a prediction of stitch height, texture, and motion, this reflects: A) Simple memorization B) A well-trained generative model that expands compressed symbols into rich predictions C) Reading ability D) Luck

3. The mental arithmetic required to place increases evenly is: A) Optional and unnecessary B) Model-based computation predicting the fabric's future shape C) Something only designers need to do D) Too difficult for crocheters

4. Reading your fabric to determine which row you are on is: A) Impossible without a row counter B) Cognitive inverse inference — using visual observations to reconstruct abstract pattern state C) Cheating D) Only possible for simple patterns

5. Recognizing a pattern repeat allows your model to: A) Skip reading the pattern entirely B) Predict the next several stitches without reading each instruction C) Work faster without accuracy D) Ignore stitch counts

6. Understanding construction logic (how pieces become garments) reflects: A) Following instructions mechanically B) Hierarchical model structure — thinking at multiple levels simultaneously C) Fashion knowledge only D) Sewing skill, not crochet skill

7. A crocheter who can look at a finished hat and reverse-engineer the pattern is using: A) Photographic memory B) Inverse inference — determining causes from observed effects using a powerful generative model C) Insider knowledge from the designer D) Simple counting

Part B: Short Answer

1. Take the instruction "Ch 4, 2 dc in 4th ch from hook, [3 dc in next ch] twice, 5 dc in last ch." Walk through the mental simulation step by step: what does your generative model predict the result will look like?
2. Explain why gauge conversion (translating between your gauge and the pattern's gauge) is a form of model-based computation. What variables must your model account for?
3. An experienced crocheter looks at a swatch of mixed stitches and can identify every stitch type. A beginner sees only bumps and loops. What has changed in the cognitive model between these two agents?